Modulation Cont'd

Intro to Audio Tech 1
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Wednesday, October 8

Georgia Tech

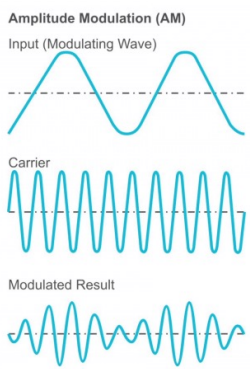
Lesson Outline

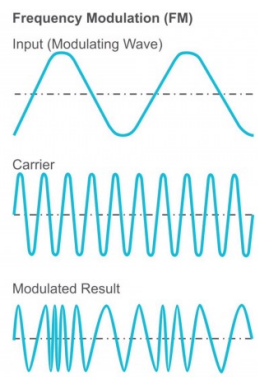
- Checkpoint 6
- Modulation Math
- Exam and Assignment Review (if time)
- Up Next…

Modulation

Modulation

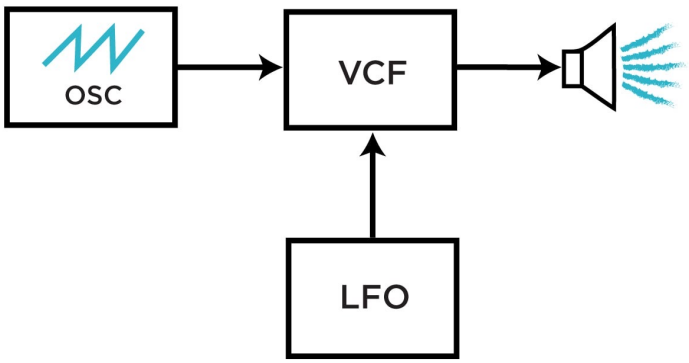
- One signal (carrier) varies due to another signal (modulator)
- The *modulator* causes the modulation and the *carrier* is what is being modulated
- Relatively inexpensive and gives us a good change over time and wide variety of timbres

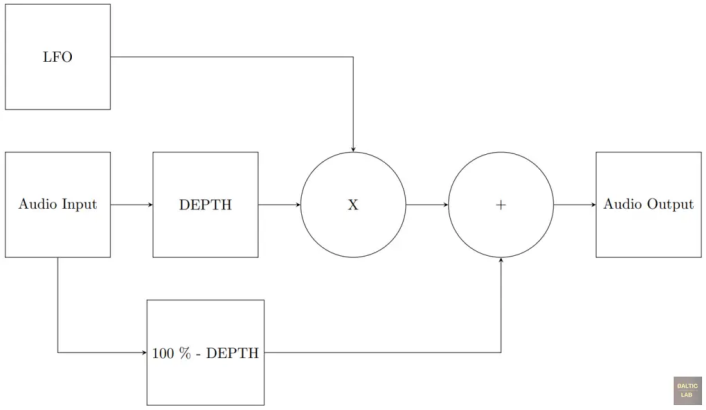




LFOs

- Low Frequency Oscillators are used to vary signals and parameters
- LFOs are sub-audio modulator signals below 20 Hz
- But modulation synthesis gets interesting when we increase the frequency of the modulator

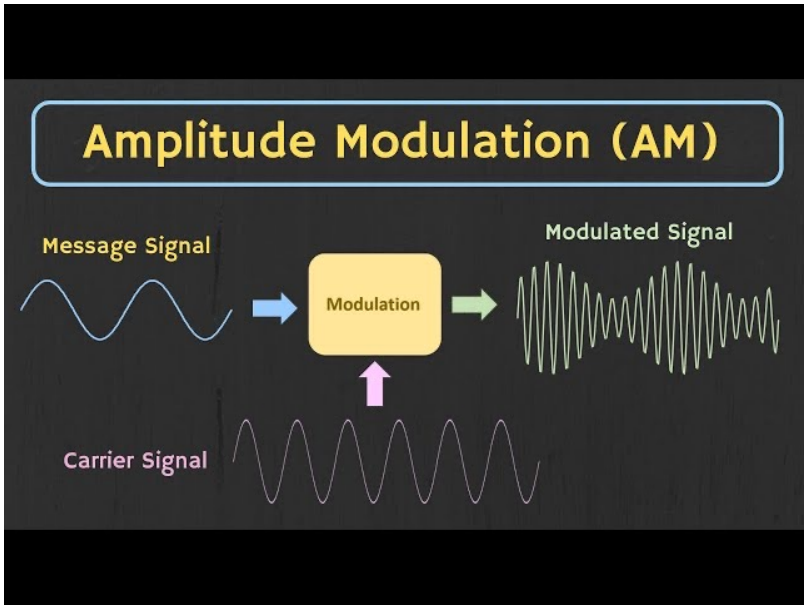




Amplitude Modulation

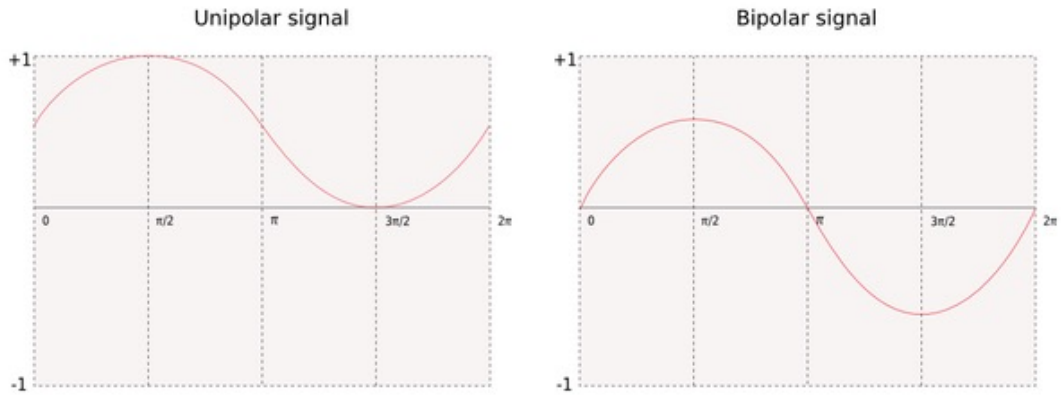
Amplitude Modulation

- Multiplying two signals together
- At slow rates (<10-20 Hz), sounds like tremolo



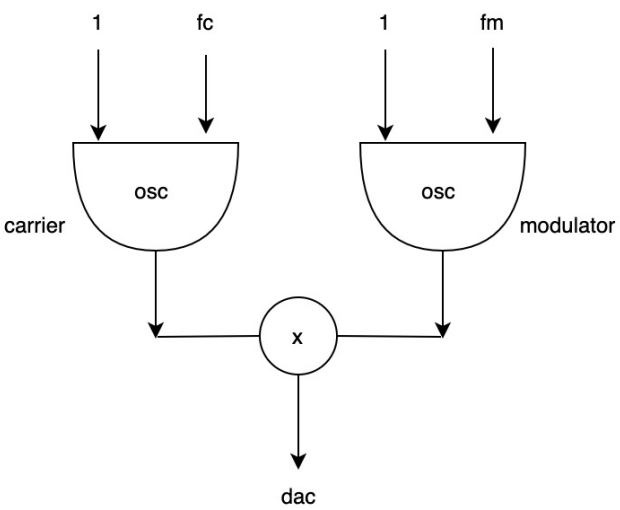
Aside: Unipolar vs Bipolar Signals

- Bipolar: positive and negative around 0
- Unipolar: shifted above 0
  - Has a DC offset



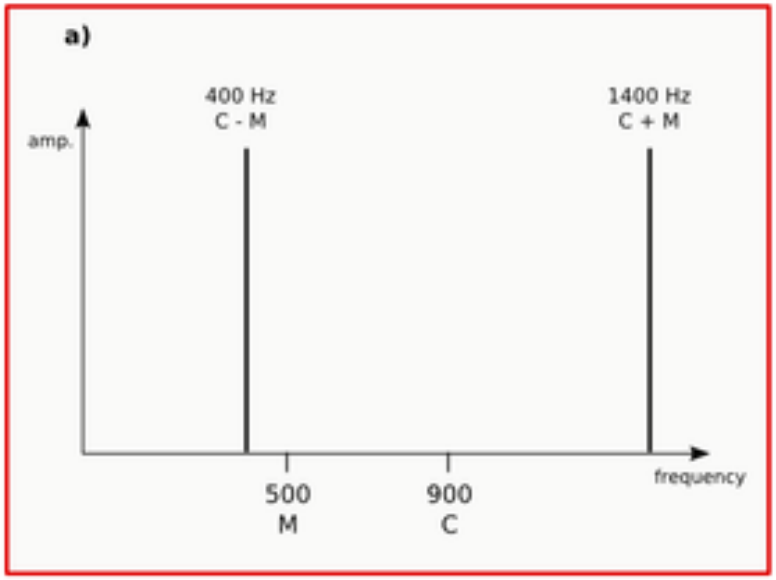
Ring Modulation

- Output = Carrier * Modulator
  - Multiply two bipolar signals in the time domain
- A modulator faster than around 10Hz causes a timbral change
- Amplitude variations are fast enough to cause **frequency sidebands**
  - Sum and difference of the carrier and modulator frequencies
- Carrier frequency is removed leaving two sidebands



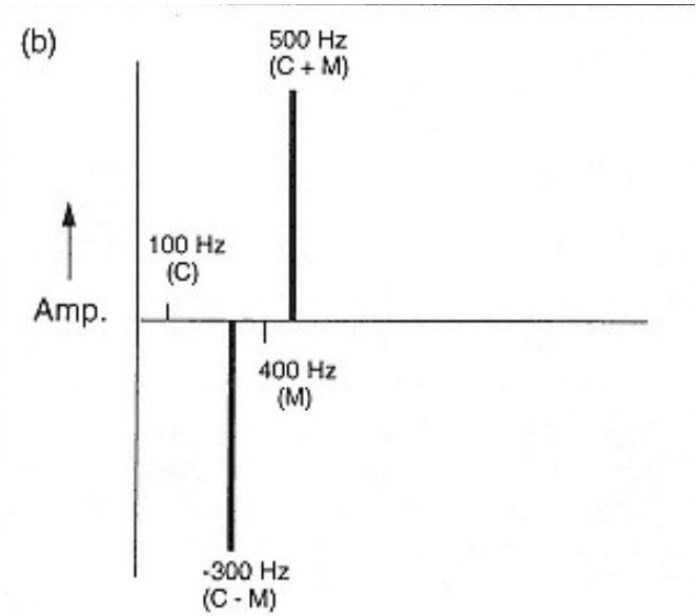
Ring Modulation

- Sinusoidal carrier at 900Hz and sinusoidal modulator at 500Hz
- Two sidebands at:
  - 900+500 = 1400Hz
  - 900-500 = 400Hz



Ring Modulation – Negative Frequencies

- Sinusoidal carrier at 100Hz and sinusoidal modulator at 400Hz
- Two sidebands at:
  - 100+400 = 500Hz
  - 100-400 = -300Hz
- Negative frequencies flip the phase



Ring Modulation

- The amplitude of the sidebands is equal to the carrier and modulator amplitudes divided by 2
- If two sine waves of amplitude 1 are modulated, what will the amplitude of the sidebands be?
  - 0.5
- We can also do ring modulation of complex or sampled signals. We'd just get more sidebands around the frequencies
- If the carrier has 3 frequency components and the modulator has 2, how many frequency components will be generated in the new signal?
  - 12 -> (3*3*2)
  - This is assuming they are not harmonically related or overlapping

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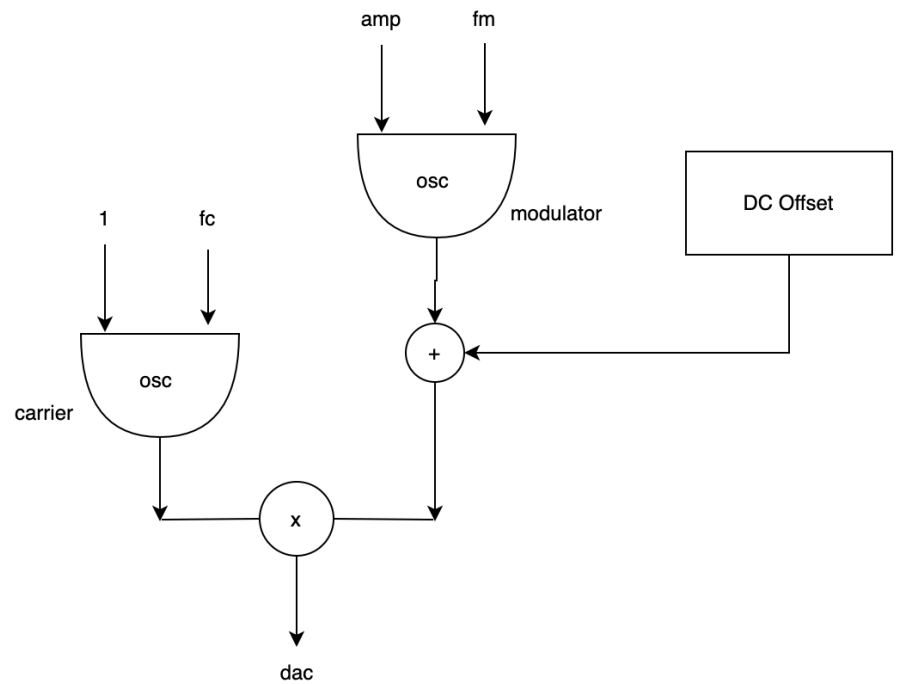
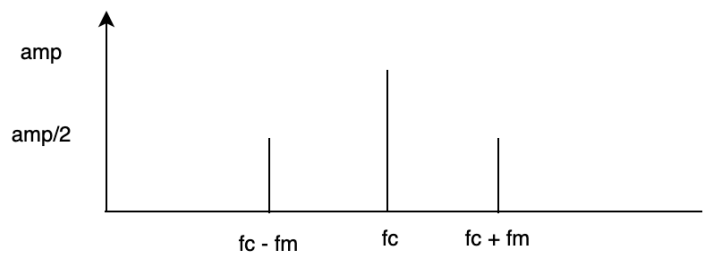
Amplitude Modulation

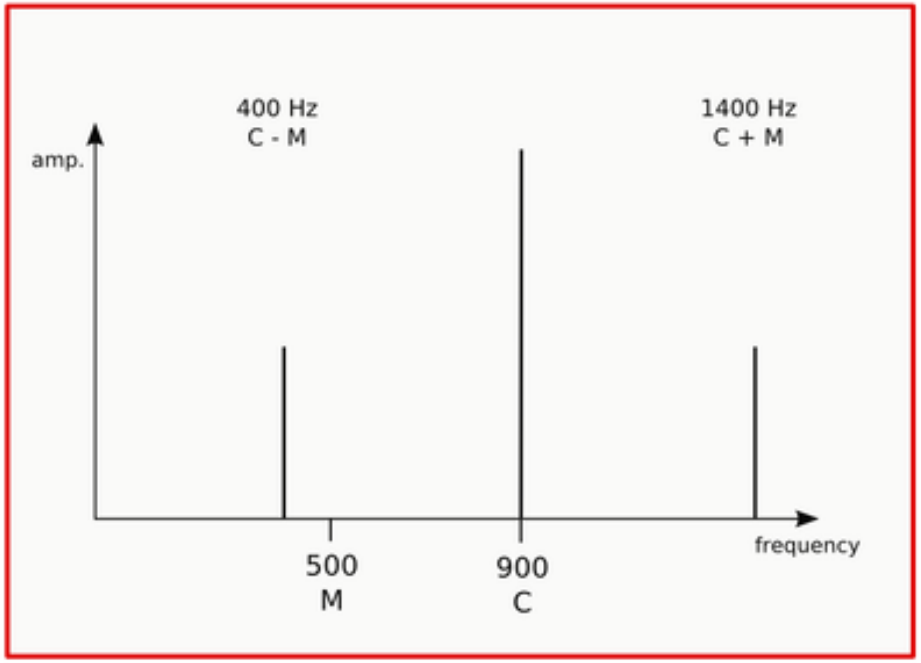
- Output = Carrier * Modulator
  - Multiply bipolar carrier and unipolar modulator
  - Preserves the carrier frequency
- Still creates sidebands at sum and difference of operator frequencies, but leaves the carrier frequency





Amplitude Modulation

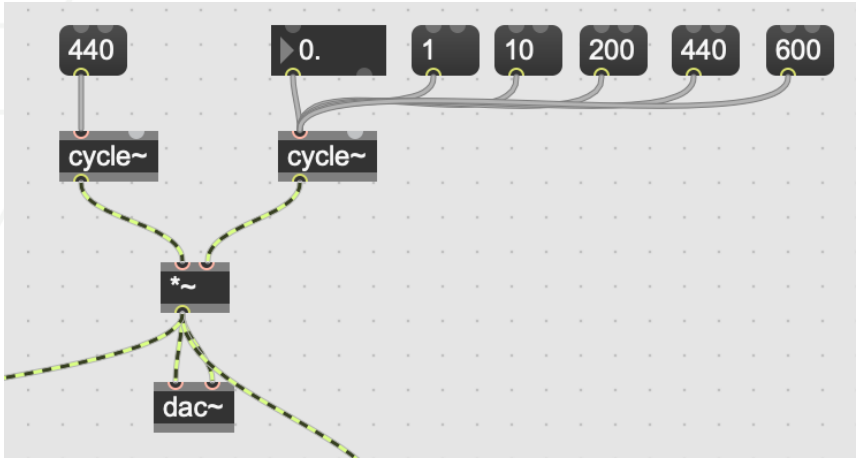
- Sinusoidal carrier at 900Hz and sinusoidal modulator at 500Hz
- Two sidebands at:
  - 900+500 = 1400Hz
  - 900-500 = 400Hz
- Sidebands are at half the amplitude of the carrier



Amplitude Modulation

- AM will create harmonic output when the carrier and modulator are harmonically related
- The “preserved” carrier frequency component will have an amplitude of the DC offset
- The sidebands will be half the amplitude of the center/carrier frequency
- You can think about the carrier being modulated by two signals the DC offset (frequency of 0 Hz) and the modulator signal

Ring Modulation in Max and Faust



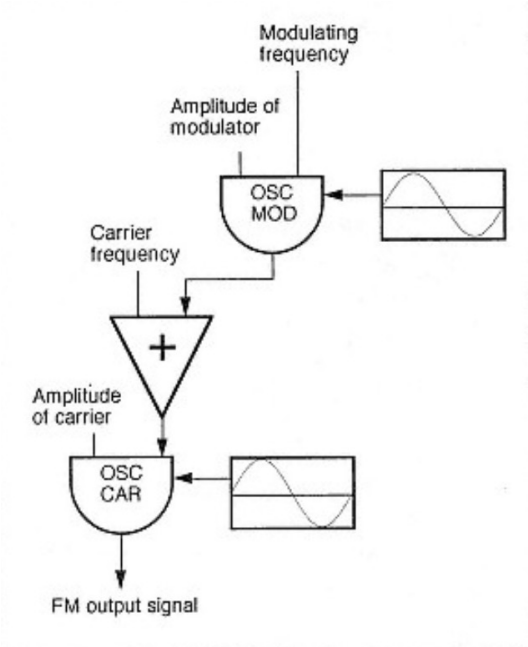

```
fc = hslider("fc", 440, 0, 1000, 1);
fm = hslider("fm", 200, 0, 300, 1);
ampc = hslider("ampc", 1, 0, 1, 0.1);
ampm = hslider("ampm", 1, 0, 1, 0.1);
carrier = os.osc(fc) * ampc;
modulator = os.osc(fm) * ampm;

process = carrier * modulator;
```

How would you turn these into amplitude modulation?

Frequency Modulation

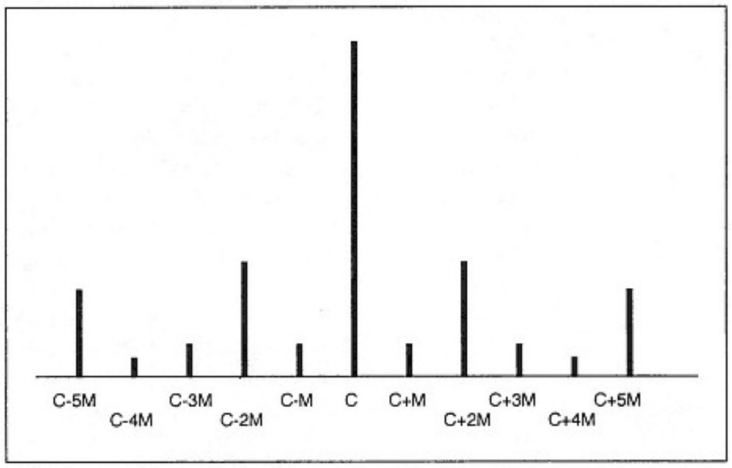
Frequency Modulation

- John Chowning at CCMRA
- Modulator signal controls the frequency of the carrier
- At low frequencies, it causes a vibrato



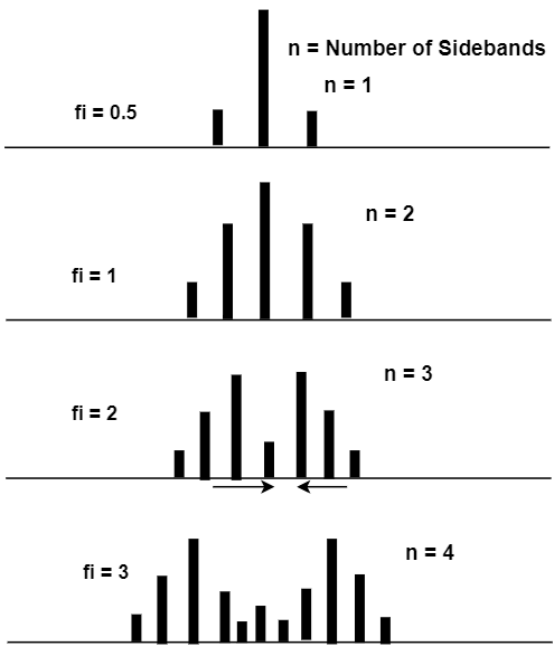
Frequency Modulation

- Creates a series of sidebands
  - Sum/difference of the carrier and multiples of the modulator
- C:M ratio
  - If the carrier and modulator are integer related, they will create harmonic sidebands



Frequency Modulation

- Theoretically, FM creates an infinite number of sidebands
- But to be perceived the sidebands need sufficient amplitude
- Depth of modulation, D(also called deviation)
  - How far from the carrier frequency is the modulator frequency M
  - The frequency deviation
  - Amplitude of the modulator
- Given the same modulator frequency, increasing the depth of modulation will increase the number of perceivable sidebands



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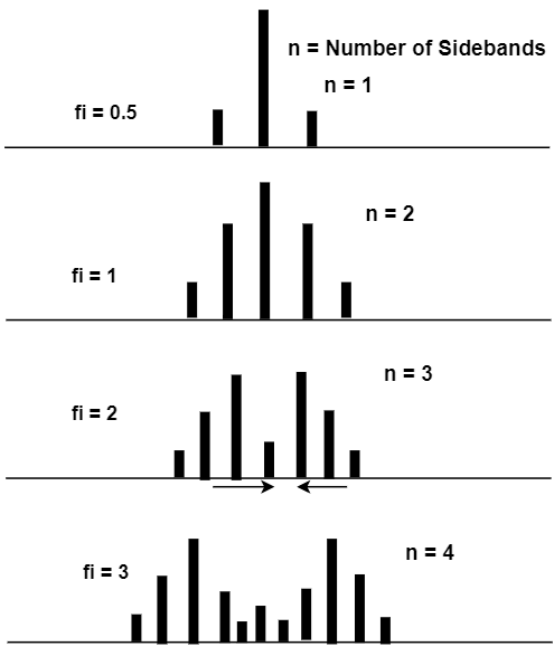
Frequency Modulation

- We can approx. the number of significant sidebands based on the relationship between the modulator frequency and depth of modulation
- Modulation index, I = D/M
- As the index increases, so do the number of sidebands
- Number of sidebands = I + 1
- So, the bandwidth of an FM signal is around 2 times the sum of the depth and modulation frequency



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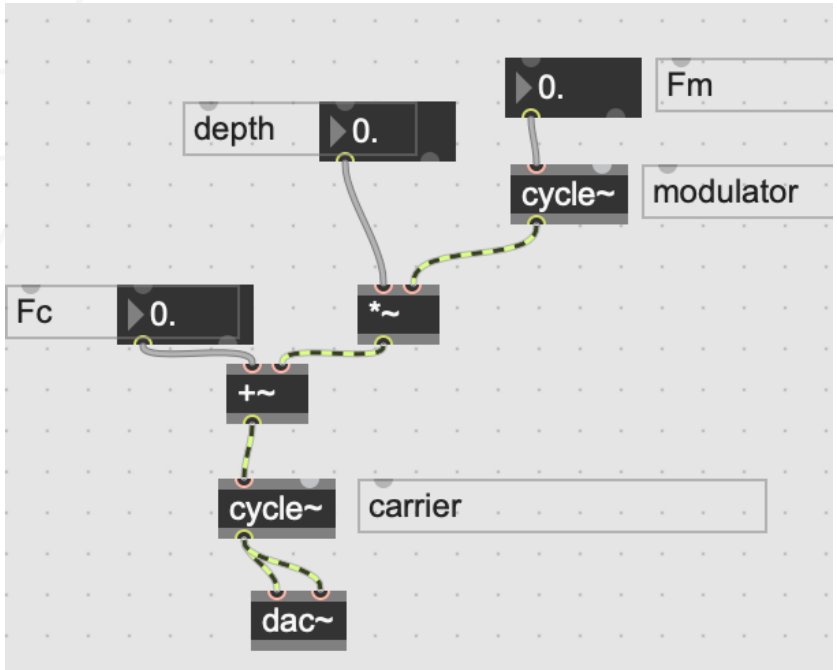
Frequency Modulation in Max and Faust




```
fc = hslider("fc", 440, 0, 1000, 1);
fm = hslider("fm", 200, 0, 300, 1);
ampc = hslider("ampc", 1, 0, 1, 0.1);
depth = hslider("depth", 1, 0, 1000, 0.5);

modulator = os.osc(fm) * depth;
carrier = os.osc(fc + modulator) * ampc;

process = carrier;
```

Up Next…

- Readings and Supplements
- Assignment 4 due next Friday 10/25 at 11:59p